

Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech

Code: 103502

Subject: Wind and Solar Energy Systems

Time: 03 Hours

Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):

[2 × 7 = 14]

- (a) The device used to convert wind energy into electrical energy is called
 - (i) Turbine
 - (ii) Transformer
 - (iii) Compressor
 - (iv) Inverter
- (b) The theoretical maximum efficiency of a wind turbine is known as
 - (i) Carnot limit
 - (ii) Rankine limit
 - (iii) Betz limit
 - (iv) Joule limit
- (c) The main material used in solar photovoltaic cells is
 - (i) Copper
 - (ii) Silicon
 - (iii) Aluminium
 - (iv) Iron
- (d) A system combining wind and solar energy sources is called
 - (i) Thermal system
 - (ii) Mechanical system
 - (iii) Hybrid renewable system
 - (iv) Hydro system
- (e) Which country is one of the largest producers of wind power in the world?
 - (i) Nepal
 - (ii) Sri Lanka
 - (iii) China
 - (iv) Pakistan
- (f) The full form of DFIG is
 - (i) Double Field Induction Generator
 - (ii) Direct Flux Induction Generator
 - (iii) Doubly Fed Induction Generator
 - (iv) Dynamic Field Integrated Generator
- (g) The converter used between generator and grid in variable-speed systems improves
 - (i) Noise level only
 - (ii) Tower height
 - (iii) Power quality
 - (iv) Blade length
- (h) Pyranometer is used to measure
 - (i) Wind speed
 - (ii) Solar radiation
 - (iii) Temperature
 - (iv) Humidity
- (i) Low Voltage Ride Through is associated with
 - (i) Battery charging
 - (ii) Water pumping
 - (iii) Wind and solar grid integration
 - (iv) Transformer cooling
- (j) A grid code specifies
 - (i) Weather conditions
 - (ii) Solar panel color
 - (iii) Technical requirements for grid connection
 - (iv) Tower height only

- Q.2** (a) Discuss the factors affecting wind power generation. [7]
(b) Explain methods for estimation of solar energy availability. [7]
- Q.3** (a) Explain the overview of grid code technical requirements for renewable energy systems. [7]
(b) Discuss isolated operation of wind-solar hybrid systems with suitable block diagram. [7]
- Q.4** (a) Explain Fresnel reflector technology with suitable diagram. [7]
(b) Explain wind speed statistics and the importance of probability distributions in wind energy assessment. [7]
- Q.5** (a) Explain the V-I characteristics of a PV cell with graph. [7]
(b) Discuss worldwide experiences of renewable energy interconnection with power systems. [7]
- Q.6** (a) Compare fixed-speed and variable-speed wind turbines. [7]
(b) Explain voltage and frequency operating limits for grid-connected renewable systems. [7]
- Q.7** (a) Discuss the advantages and disadvantages of parabolic trough systems. [7]
(b) Describe the construction and operation of PV module and PV array. [7]
- Q.8** (a) Explain different solar thermal energy technologies with neat diagrams. [7]
(b) Explain Maximum Power Point Tracking and its importance. [7]
- Q.9** Write short note on *any two* of the following: [7 x 2 = 14]
(a) Explain Polycrystalline silicon
(b) Explain solar pond
(c) What is air density
(d) Define grid synchronization
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